Supplementary Material for "MixPTS: Mixture of Endpoints for Diffusion-based Probabilistic Time Series Forecasting"

This document complements the paper with the derivations behind Section IV-A, the complete experimental configuration, and additional results that the page limit of the paper does not accommodate. Equation, table and figure numbers without the prefix S refer to the paper.

S1. Derivations

A. Proof of Proposition 1

Let $p = \sum_k \rho_k \mathcal{N}(\mu_k, \sigma_k^2)$ and write $\bar{\mu} = \sum_k \rho_k \mu_k$. For a Gaussian $\mathcal{N}(f, g)$ with $g > 0$, the expected negative log-likelihood under p equals

$$\ell(f, g) = \tfrac{1}{2}\log(2\pi g) + \frac{\mathbb{E}_{y\sim p}\big[(y-f)^2\big]}{2g}. \tag{S1}$$

For fixed g, the numerator is minimized by $f^\star = \mathbb{E}_p[y] = \bar{\mu}$, and by the law of total variance its minimum equals

$$V = \mathrm{Var}_p(y) = \sum_k \rho_k \sigma_k^2 + \sum_k \rho_k (\mu_k - \bar{\mu})^2. \tag{S2}$$

The remaining function $\frac{1}{2}\log g + V/(2g)$ has the derivative $(g - V)/(2g^2)$, which is negative for $g < V$ and positive for $g > V$, so that the unique minimizer is $g^\star = V$. This establishes Eq. (4). In a regime with variance σ_j^2, the reported variance therefore deviates from the true one by $g^\star - \sigma_j^2$; for two regimes with $\sigma_1 < \sigma_2$ this deviation equals $\rho_2(\sigma_2^2 - \sigma_1^2) + \sum_k \rho_k(\mu_k - \bar{\mu})^2 > 0$ in the calm regime and $-\rho_1(\sigma_2^2 - \sigma_1^2) + \sum_k \rho_k(\mu_k - \bar{\mu})^2$ in the volatile one, which are the two expressions stated in the paper. Because the mixture family contains every single Gaussian, the optimal expected likelihood of a K-component mixture is never worse than that of $\mathcal{N}(f^\star, g^\star)$, and for the two-regime law it attains the entropy of p itself.

B. Excess CRPS of a Mis-scaled Gaussian

Let $Z, Z' \sim \mathcal{N}(\mu, s^2)$ be independent draws from the forecast and let $y \sim \mathcal{N}(\mu, \sigma^2)$ be independent of both. Using the kernel form of the CRPS in Eq. (1) and the identity $\mathbb{E}|W| = \sqrt{2/\pi}\,\omega$ for $W \sim \mathcal{N}(0, \omega^2)$, the two expectations become $\mathbb{E}|Z - y| = \sqrt{2/\pi}\,\sqrt{s^2 + \sigma^2}$ and $\mathbb{E}|Z - Z'| = 2s/\sqrt{\pi}$, so that

$$\mathbb{E}\big[\mathrm{CRPS}(\mathcal{N}(\mu, s^2), y)\big] = \sqrt{\tfrac{2}{\pi}}\sqrt{s^2 + \sigma^2} - \frac{s}{\sqrt{\pi}}. \tag{S3}$$

At $s = \sigma$ this expression equals $\sigma/\sqrt{\pi}$, and subtracting this value yields the excess $\Delta(s; \sigma)$ of Eq. (5). The excess is non-negative because $\sqrt{2(s^2 + \sigma^2)} \geq s + \sigma$ by the inequality between the quadratic and the arithmetic mean, with equality if and only if $s = \sigma$. For the example of the paper, two equally likely regimes with $\sigma_2 = 3\sigma_1$ and equal means, Eq. (S2) gives $s = \sqrt{5}\,\sigma_1$, and the relative excess $\Delta(s; \sigma_j)\sqrt{\pi}/\sigma_j$ equals $\sqrt{12} - \sqrt{5} - 1 = 0.228$ in the calm regime and $(\sqrt{28} - \sqrt{5} - 3)/3 = 0.018$ in the volatile one.

The central interval of level ℓ of $\mathcal{N}(\mu, s^2)$ is $\mu \pm s\, z_\ell$ with $z_\ell = \Phi^{-1}\big((1 + \ell)/2\big)$, and its coverage under $\mathcal{N}(\mu, \sigma^2)$ equals $2\Phi(z_\ell\, s/\sigma) - 1$. With $s^2 = g^\star$, this expression produces the coverage curves of Fig. 3(b); in the example above, the nominal 50% interval covers 86.8% of the calm targets and 38.5% of the volatile ones, which reproduces the over-coverage of calm windows that Fig. 1(b) measures for DiffPTS.

C. Gradient of the Mixture Likelihood with Respect to the Gate

Write $\mathcal{N}_{k,m,n} = \mathcal{N}\big(y_{m,n}; f_{k,m,n}(X), g_{k,m,n}(X)\big)$ and $\pi_{n,k} = \mathrm{softmax}_k(z_{n,k}/\tau)$. The posterior responsibility of expert k for entry (m, n) is $r_{m,n,k} = \pi_{n,k}\mathcal{N}_{k,m,n}/\sum_j \pi_{n,j}\mathcal{N}_{j,m,n}$, and differentiating Eq. (11) through the softmax yields

$$\frac{\partial \mathcal{L}_{\mathrm{mix}}}{\partial z_{n,k}} = -\frac{1}{MN\tau}\sum_{m=1}^{M}\big(r_{m,n,k} - \pi_{n,k}\big). \tag{S4}$$

A descent step therefore raises the logit of every expert whose average responsibility over the horizon exceeds its current weight, and the step size scales with $1/\tau$. The joint alternative discussed in Section IV-F replaces the entry-wise responsibilities by a single responsibility per variable, $r_{n,k} \propto \pi_{n,k}\prod_m \mathcal{N}_{k,m,n}$, whose log-odds between two experts are sums of M terms; for a horizon of $M = 192$ steps these responsibilities are almost always binary, so that the gate receives a gradient from only one expert per variable and window. The last rows of Table S6 quantify the consequence.

S2. Experimental Details

A. Architecture and Training

Conditioning trunk. The trunk follows the reference implementation of each dataset: PatchTST [1] with patches of length 16 and stride 8 for the four ETT datasets, and the non-stationary Transformer [2] for the other five, both with model width 512, eight attention heads, two encoder layers, a feed-forward width of 1,024 and dropout 0.05.

Experts and gate. Each residual head h_k maps the horizon of a variable through $\mathbb{R}^M \to \mathbb{R}^{256} \to \mathbb{R}^M$ with a GELU activation. The variance experts share an embedding φ that

TABLE S1
CONFIGURATION OF THE TRAINED MIXPTS MODELS BEHIND TABLE III. ENTRIES SEPARATED BY SLASHES GIVE THE VALUES OF SEEDS 1, 2 AND 3; A SINGLE ENTRY APPLIES TO ALL THREE SEEDS TRAINED ON THAT DATASET.

Dataset	K	τ	λ	Trunk
ETTh1	2/4/4	0.5	10^{-3}	PatchTST
ETTh2	2	1	10^{-2}	PatchTST
ETTm1	4/2/2	0.5	10^{-3}	PatchTST
ETTm2	2	0.25/0.25/0.5	$10^{-2}/10^{-2}/10^{-3}$	PatchTST
ECL	4	0.5	10^{-3}	NS-Transformer
EXG	4	0.5	10^{-3}	NS-Transformer
ILI	4	0.5	10^{-3}	NS-Transformer
Solar	4	0.25/0.5/0.5	$10^{-2}/10^{-3}/10^{-3}$	NS-Transformer
Traffic	4	1/0.5/0.5	10^{-3}	NS-Transformer

maps the look-back window of a variable to $\mathbb{R}^{512}$ with a GELU activation, followed by one linear output layer per expert, a softplus and $\epsilon = 10^{-7}$. The gate is a perceptron with two hidden layers of 64 units and GELU activations that maps the $6N$ statistics of Eq. (9) to $N \cdot K$ logits.

Diffusion and optimization. The denoiser and the location-scale reverse process are those of the reference implementation, with $T = 20$ steps and a linear variance schedule from 10^{-4} to 10^{-2}, which is also the schedule of the released DiffPTS code. All parameters are optimized jointly with Adam at a learning rate of 10^{-4}, an effective batch size of 32 (some models accumulate two mini-batches of 16 to fit into memory), gradient clipping at a norm of 5, at most ten epochs and early stopping with a patience of five epochs on the validation CRPS; the checkpoint with the lowest validation CRPS is evaluated on the test split. Sampling draws $S = 100$ trajectories per window in parallel chunks.

Configuration of the trained models. Table S1 lists the number of experts, the gate temperature and the balance weight of every trained model behind Table S2; all remaining settings are shared by all datasets.

B. Evaluation

Protocol scores. Tables III, IV and V and Tables S2, S3 and S6 use the evaluation of the reference implementation, which scores the non-overlapping windows of each test split with 100 trajectories and averages the CRPS and the MSE over windows, horizon steps and variables on the standardized scale; for every run, we report the test evaluation of the epoch with the lowest validation CRPS.

Dense same-window evaluation. The paired analyses of Sections V-C–V-F instead score a regular subsample of the overlapping sliding windows, namely every fourth window on ETTh1 and ETTh2, every sixteenth on ETTm1, ETTm2 and SolarEnergy, every second on ExchangeRate and every window on ILI. Both methods are evaluated on the same windows with the same sampling seed, and the identity of the windows is verified through the per-window sums of the targets, which agree exactly. The scores of a method are averaged over its trained models before two methods are compared, and one SolarEnergy model and one Electricity model, whose stored weights do not reproduce their logged test scores, are left out of every analysis of trained models. Confidence intervals for the mean per-window difference come from a circular moving-block bootstrap [3] with 4,000 resamples and blocks that cover $\lceil (C+M)/\text{stride} \rceil$ consecutive windows, so that windows sharing an observation fall into the same block; on ILI the 162 windows form a single block, and no interval is reported.

Re-executed baselines. DiffPTS [4] and NsDiff [5] were re-executed with their official code, scripts and seeds on the seven datasets for which three completed runs of both are available; these models also serve as the single-Gaussian reference of the paired analyses.

C. Code

The code of MIXPTS is available at https://github.com/wwy155/MixPTS. Every experiment runs on a single NVIDIA RTX 3090 GPU.

S3. ADDITIONAL RESULTS

A. Per-Seed Results and Re-executed Baselines

Table S2 lists the test scores of every seed behind Table III. The standard deviations across seeds are small compared with the margins of the main comparison on most datasets, and the largest spread occurs on ILI, whose test split contains a single non-overlapping window. Table S3 lists the per-seed CRPS of DiffPTS and NsDiff re-executed under identical conditions, whose means appear in Table IV. MIXPTS attains the lower CRPS and the lower MSE on all seven datasets, and its margins over DiffPTS exceed those of Table III on ETTh2, ETTm2, ExchangeRate and ILI, where the re-executed DiffPTS is less accurate than reported; the re-executed NsDiff is less accurate than reported on all seven datasets.

B. Dense Evaluation on All Seven Datasets

Table S4 extends the paired comparison of Fig. 4(a) to all seven datasets with re-executed baselines and adds the calibration diagnostics of Section V-D. On the four ETT datasets and on SolarEnergy, the dense evaluation confirms the protocol results. On ExchangeRate, the mixture produces intervals that are 16% narrower than those of DiffPTS, which lowers the MSE by 4.3% but leaves the nominal 90% interval below its target, so that the dense CRPS of the two models is statistically indistinguishable. ILI is the shortest series of the benchmark, with 966 weekly observations, and both models under-cover the test period severely; there, the dense evaluation favours the single Gaussian by 2.1%, whereas the protocol score of Table III, which rests on one test window, favours the mixture.

Fig. S1 decomposes the paired CRPS reduction of Section V-C by volatility. On all four ETT datasets, the reduction is largest in the calmer groups and shrinks toward the volatile end: it reaches 24% in the calmest fifth of the ETTm1 windows, 19% and 21% in the second-calmest fifth of ETTh2 and ETTm2, and 6.5% in the calmest fifth of ETTh1, which is the pattern of Fig. 3(a). SolarEnergy exhibits the complementary pattern, with the largest reduction (13%) in the most volatile fifth of the future windows, where DiffPTS under-covers.

TABLE S2
PER-SEED TEST SCORES OF MIXPTS WITH MEAN AND STANDARD DEVIATION OVER THE THREE SEEDS.

	CRPS				MSE			
Dataset	seed 1	seed 2	seed 3	mean±sd	seed 1	seed 2	seed 3	mean±sd
ETTh1	.3549	.3467	.3544	.352±.005	.4660	.4676	.4666	.467±.001
ETTh2	.3564	.3597	.3470	.354±.007	.4264	.4716	.4371	.445±.024
ETTm1	.3037	.2996	.2958	.300±.004	.3784	.3773	.3735	.376±.003
ETTm2	.2476	.2579	.2610	.256±.007	.2363	.2640	.2812	.261±.023
ECL	.2156	.2150	.2154	.215±.000	.1992	.1962	.1997	.198±.002
EXG	.2682	.2798	.2601	.269±.010	.2120	.2304	.1934	.212±.018
ILI	.7466	.7550	.6785	.727±.042	2.5924	2.4833	1.9682	2.348±.333
Solar	.1701	.1670	.1630	.167±.004	.2014	.2016	.1924	.198±.005
Traffic	.2662	.2628	.2608	.263±.003	.6272	.6266	.6102	.621±.010

TABLE S3
PER-SEED TEST CRPS (SEEDS 1 / 2 / 3) AND THREE-SEED MEANS OF THE TWO STRONGEST BASELINES RE-EXECUTED WITH THEIR OFFICIAL CODE AND SEEDS; THE MEANS APPEAR IN TABLE IV OF THE MAIN PAPER.

	NsDiff		DiffPTS	
Dataset	seeds	mean	seeds	mean
ETTh1	.4534 / .4208 / .4371	.437	.3659 / .3711 / .3605	.366
ETTh2	.4789 / .4321 / .4548	.455	.3721 / .3740 / .3792	.375
ETTm1	.3450 / .3437 / .3552	.348	.3267 / .3160 / .3122	.318
ETTm2	.3787 / .3921 / .3840	.385	.2902 / .2908 / .2962	.292
EXG	.4575 / .3235 / .4464	.409	.2938 / .3142 / .2644	.291
ILI	.9343 / .8752 / .8994	.903	.8152 / .8121 / .7803	.803
Solar	.3599 / .3827 / .3804	.374	.1738 / .1679 / .1693	.170

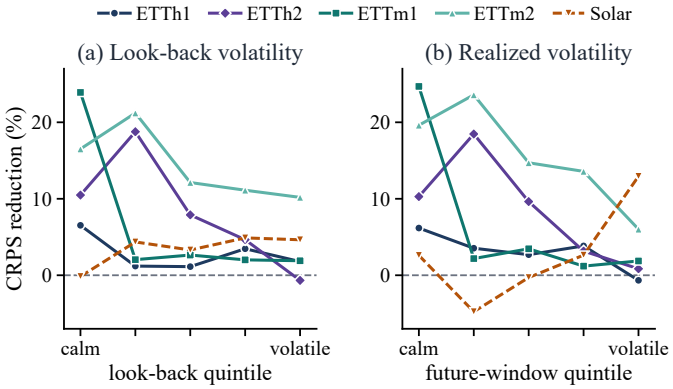


Fig. S1. Paired CRPS reduction of MIXPTS over DiffPTS in five groups of (window, variable) pairs, ordered (a) by the volatility of the look-back window and (b) by the realized volatility of the future window.

C. Trained Experts and Gates

Table S5 reports, for every trained model analysed in Section V-E, the largest average expert weight, the normalized gate entropy, the ratio between the widest and the narrowest expert, and the rank correlations between look-back volatility and the routed and the unweighted expert width. No model collapses onto a single expert: the largest average weight never exceeds 0.78, and the normalized entropy lies between 0.45 and 0.99.

D. Controlled Variants per Seed

Table S6 lists the per-seed test CRPS of the controlled variants whose means appear in Fig. 7 and Table V, together with the gate temperatures and balance weights behind the robustness statement of Section V-F. On SolarEnergy, the per-variable gate attains .1702, .1670 and .1630 for the three seeds, against .2676, .2853 and .3026 for the per-window gate.

E. Design Ablations

The last three rows of Table S6 replace one design choice of Section IV-F at a time and keep every other setting of the four-expert model. Conditioning the denoiser on the gate-weighted average of the expert means and variances, instead of on one selected expert, raises the mean CRPS by 1.3% on ETTh1 and by 3.8% on ETTm1. Evaluating the mixture likelihood jointly over the horizon of each variable raises it by 8.9% and 14.4%, and on ETTm1 this variant is even less accurate than a single expert, which is consistent with the nearly binary responsibilities discussed in Section S1. Finally, a gate that reads the standardized look-back window of every variable instead of the six statistics of Eq. (9) raises the CRPS by 13.3% and 9.5%, although it has access to strictly more information. Each of the three choices therefore contributes to the accuracy of MIXPTS.

F. Multivariate Scores

Since the gate selects an expert for every variable independently, we also score the joint forecast of all variables on the dense test windows, with the same trained models, sampling seed and 100 trajectories as in Section V-C. The energy score $\mathrm{ES} = \frac{1}{S}\sum_s \|x_s - y\| - \frac{1}{2S^2}\sum_{s,s'} \|x_s - x_{s'}\|$ is computed on the vector of all variables at every horizon step and averaged over the horizon [6], and CRPS-sum is the CRPS of the sum over all variables. Table S7 lists both scores. On the four ETT datasets, MIXPTS lowers the energy score of DiffPTS by 2.3–11.7% with intervals that exclude zero, and it lowers CRPS-sum on ETTh1, ETTh2 and ETTm2, whereas on ETTm1 the two models are tied in CRPS-sum. On ExchangeRate both differences are within their intervals, and on ILI and SolarEnergy the energy score of MIXPTS is slightly higher, by 1.8% and 1.0%.

TABLE S4

DENSE SAME-WINDOW COMPARISON OF MIXPTS (M) WITH RE-EXECUTED DIFFPTS (D) ON THE SEVEN DATASETS WITH THREE RE-EXECUTED SEEDS. W: NUMBER OF EVALUATED WINDOWS; GAIN: RELATIVE CRPS REDUCTION OF MIXPTS WITH A 95% BLOCK-BOOTSTRAP INTERVAL; WIN: SHARE OF WINDOWS ON WHICH MIXPTS HAS THE LOWER CRPS; ACE: AVERAGE COVERAGE ERROR; w_{90}: WIDTH RATIO OF THE 90% INTERVALS (M/D); cov_{90}: EMPIRICAL COVERAGE OF THE NOMINAL 90% INTERVAL.

		CRPS					ACE			cov_{90}	
Dataset	W	M	D	gain (%)	95% CI	win (%)	M	D	w_{90}	M	D
ETTh1	673	.3434	.3530	+2.7	[1.6, 3.8]	82	.080	.142	0.94	.935	.960
ETTh2	673	.3444	.3692	+6.7	[1.6, 12.1]	76	.091	.139	0.95	.918	.941
ETTm1	709	.2954	.3129	+5.6	[4.8, 6.3]	92	.036	.058	0.93	.899	.908
ETTm2	709	.2645	.3046	+13.2	[11.6, 14.6]	93	.110	.171	0.90	.942	.956
EXG	663	.2766	.2699	−2.5	[−10.1, 4.0]	52	.168	.049	0.84	.691	.860
ILI	162	.8573	.8395	−2.1	–	32	.287	.246	0.91	.424	.459
Solar	646	.1810	.1868	+3.1	[0.6, 5.7]	58	.063	.061	1.15	.857	.790

TABLE S5

EXPERTS AND GATES OF THE TRAINED MODELS. $\max \bar{\pi}$: LARGEST AVERAGE EXPERT WEIGHT; $H/\log K$: NORMALIZED ENTROPY OF THE GATE; RATIO: WIDEST OVER NARROWEST AVERAGE EXPERT WIDTH; ρ: SPEARMAN CORRELATION BETWEEN LOOK-BACK VOLATILITY AND THE ROUTED / UNWEIGHTED WIDTH (COMPUTED ONLY ON THE DATASETS OF THE PAIRED COMPARISON).

Dataset	seed	K	$\max \bar{\pi}$	$H/\log K$	ratio	ρ routed / unw.
ETTh1	1	2	0.55	0.92	1.40	−0.09 / −0.07
ETTh1	2	4	0.34	0.54	3.04	0.28 / −0.77
ETTh1	3	4	0.36	0.76	1.61	0.22 / −0.46
ETTh2	1	2	0.66	0.85	1.77	0.03 / 0.29
ETTh2	2	2	0.51	0.99	1.00	0.47 / 0.48
ETTh2	3	2	0.56	0.97	1.05	0.44 / 0.44
ETTm1	1	4	0.34	0.53	2.45	0.53 / −0.10
ETTm1	2	2	0.52	0.63	2.32	0.57 / −0.41
ETTm1	3	2	0.53	0.61	1.99	0.56 / −0.19
ETTm2	1	2	0.67	0.79	4.36	0.33 / 0.55
ETTm2	2	2	0.61	0.86	1.49	0.54 / 0.52
ETTm2	3	2	0.57	0.85	3.23	0.47 / 0.54
EXG	1	4	0.44	0.45	7.54	0.10 / −0.21
EXG	2	4	0.41	0.71	2.54	0.03 / −0.18
EXG	3	4	0.42	0.55	5.65	0.03 / −0.23
ILI	1	4	0.31	0.90	1.39	−0.42 / −0.45
ILI	2	4	0.75	0.53	2.63	−0.33 / −0.40
ILI	3	4	0.49	0.67	2.37	−0.38 / −0.43
Solar	1	4	0.33	0.87	1.11	0.28 / 0.37
Solar	3	4	0.34	0.68	1.96	0.20 / 0.10
ECL	2	4	0.51	0.60	4.36	–
ECL	3	4	0.41	0.61	5.94	–
Traffic	1	4	0.66	0.61	5.07	–
Traffic	2	4	0.69	0.56	4.64	–
Traffic	3	4	0.78	0.46	3.70	–

TABLE S6

PER-SEED TEST CRPS (SEEDS 1 / 2 / 3) OF THE CONTROLLED VARIANTS. ONLY THE LISTED FACTOR DIFFERS FROM THE FOUR-EXPERT MODEL.

Variant	ETTh1	ETTm1
One expert	.3785 / .3840 / .3903	.3108 / .3081 / .3173
One wide expert	.3660 / .3804 / .3846	.3071 / .3184 / .3164
Two experts	.3553 / .3647 / .3557	.2921 / .2996 / .2958
Four experts	.3469 / .3447 / .3557	.3037 / .2896 / .3057
Four, per-window gate	.3689 / .3764 / .3783	.3416 / .3503 / .3290
Four, $\tau = 0.25$	.3498 / .3478 / .3787	–
Four, $\tau = 1$	.3594 / .3435 / .3595	–
Four, $\lambda = 0$	.3557 / .3409 / .3548	–
Four, $\lambda = 10^{-2}$	.3419 / .3509 / .3461	–
Four, averaged path	.3544 / .3539 / .3529	.3077 / .3105 / .3148
Four, joint likelihood	.3768 / .3871 / .3769	.3386 / .3613 / .3281
Four, raw-window gate	.4014 / .3738 / .4117	.3181 / .3477 / .3185

G. Endpoint-Only Forecasts

Section V-F scores the endpoint distributions of both models without the reverse process. For an endpoint $\sum_k w_k \mathcal{N}(\mu_k, s_k^2)$ and an observation y, the CRPS has the closed form [7]

$$\mathrm{CRPS} = \sum_k w_k A\big(y - \mu_k, s_k^2\big) - \tfrac{1}{2} \sum_{k,l} w_k w_l A\big(\mu_k - \mu_l, s_k^2 + s_l^2\big),$$

where $A(\mu, s^2) = \mu\,(2\Phi(\mu/s) - 1) + 2s\,\varphi(\mu/s)$, and the Gaussian endpoint of DiffPTS is the case of a single component. A Monte Carlo estimate with 4,000 draws per entry agrees with the closed form to within 0.05% on ETTh1 and ETTm1. Coverage follows from the PIT value of every observation under the endpoint, and the windows, the trained models and the averaging are those of the dense evaluation. Table S8 lists the results.

The mixture endpoint has a lower CRPS than the Gaussian endpoint on six of the seven datasets and a lower coverage error on the four ETT datasets and on SolarEnergy; on ILI its CRPS is 1.0% higher, and on ExchangeRate its coverage error is higher. On the four ETT datasets, moreover, the endpoint of each model scores lower than the corresponding sampler with 100 trajectories, by 4.4–10.2% for DiffPTS and by 2.9–7.0% for MIXPTS, and the reverse process widens the central 90% interval of DiffPTS by 3–31% on these datasets. This observation concerns the reverse process of the reference protocol, which both models share, and it agrees with the recent finding that the final denoising steps of diffusion forecasters can degrade their forecasts [8]. Improved samplers are therefore complementary to the mixture endpoint. On SolarEnergy, in contrast, the reverse process lowers the CRPS of the Gaussian endpoint by 5.8%.

H. Post-hoc Recalibration

Section V-G compares MIXPTS with recalibrated DiffPTS models. All recalibration maps are fitted on the validation split, using every window at the stride of the dense test evaluation, the same sampling seed and 100 trajectories, and they are then applied unchanged to the test windows. *Global rescaling* multiplies the deviation of every sample from the sample mean

TABLE S7

MULTIVARIATE SCORES ON THE DENSE TEST WINDOWS (MEANS OVER THE TRAINED MODELS; LOWER IS BETTER). D: DIFFPTS; M: MIXPTS; GAIN: RELATIVE REDUCTION BY MIXPTS IN PERCENT WITH A 95% BLOCK-BOOTSTRAP INTERVAL. ILI HAS A SINGLE BLOCK, SO ITS INTERVAL IS DEGENERATE.

	CRPS-sum				Energy score			
Dataset	D	M	gain (%)	95% CI	D	M	gain (%)	95% CI
ETTh1	1.452	1.410	+2.9	[0.8, 4.7]	1.174	1.147	+2.3	[1.3, 3.2]
ETTh2	1.620	1.441	+11.0	[3.5, 18.1]	1.221	1.139	+6.7	[2.4, 11.1]
ETTm1	1.252	1.247	+0.4	[−0.9, 1.7]	1.025	.9819	+4.2	[3.5, 4.9]
ETTm2	1.393	1.231	+11.6	[9.4, 13.6]	1.032	.9116	+11.7	[10.3, 13.0]
EXG	1.459	1.375	+5.8	[−16.2, 24.7]	.9222	.9153	+0.7	[−6.1, 6.5]
ILI	5.100	5.236	−2.7	[−2.7, −2.7]	2.577	2.624	−1.8	[−1.8, −1.8]
Solar	27.945	27.849	+0.3	[−3.9, 4.6]	2.601	2.627	−1.0	[−2.8, 1.0]

TABLE S8

ENDPOINT-ONLY FORECASTS, SCORED IN CLOSED FORM, AGAINST THE SAMPLERS WITH 100 TRAJECTORIES ON THE DENSE TEST WINDOWS (MEANS OVER THE TRAINED MODELS). END: ENDPOINT DISTRIBUTION; FULL: REVERSE DIFFUSION. BOLD MARKS THE BEST CRPS AND THE BEST ACE OF EACH ROW.

	DiffPTS				MixPTS			
	CRPS		ACE		CRPS		ACE	
Dataset	end	full	end	full	end	full	end	full
ETTh1	.3310	.3530	.067	.142	**.3277**	.3434	**.013**	.080
ETTh2	.3316	.3692	.131	.139	**.3202**	.3444	**.068**	.091
ETTm1	.2992	.3129	.054	.058	**.2867**	.2954	**.027**	.036
ETTm2	.2769	.3046	.185	.171	**.2475**	.2645	**.103**	.110
EXG	.2825	**.2699**	.059	**.049**	.2779	.2766	.210	.168
ILI	**.8307**	.8395	.287	**.246**	.8389	.8573	.314	.287
Solar	.1983	.1868	.235	**.061**	**.1778**	.1810	.082	.063

by one factor a, chosen from the grid $\{0.40, 0.45, \ldots, 1.80\}$ to minimize the validation CRPS. *Volatility-aware rescaling* splits the (window, variable) pairs into five groups by the quintiles of the look-back standard deviation on the validation split and chooses one factor per group. *Isotonic recalibration* [9] maps the quantile levels of the samples through the inverse of the empirical distribution of the randomized PIT on the validation split, either with one map per model or with one map per variable. Table S10 reports the resulting test CRPS for both models and Table S9 the factors of the volatility-aware map.

On the four ETT datasets and on SolarEnergy, every rescaling map leaves MIXPTS ahead of DIFFPTS when both models receive the same map, and on ETTh2, ETTm1, ETTm2 and SolarEnergy the raw MIXPTS is more accurate than DIFFPTS with volatility-aware rescaling. The factors of DIFFPTS increase monotonically with the volatility quintile on ETTm1 and ExchangeRate and cross one, which is the two-sided error of Proposition 1; on ETTh1, ETTh2 and ETTm2 they rise from 0.40–0.55 in the calmest quintile to 0.73–0.93 in the most volatile one, so that calm windows require the strongest narrowing. Isotonic recalibration is the most flexible map and helps both models on ETTh2 and ETTm2; with one map per variable it gives DIFFPTS a slightly lower CRPS than MIXPTS on ETTh1 (.3364 against .3404). On ExchangeRate and ILI, whose test periods differ strongly from their validation periods, the maps fitted on validation hardly transfer: apart from a 0.4% gain of DIFFPTS with volatility-aware rescaling on ExchangeRate, they increase the test CRPS of both models.

TABLE S9

SPREAD FACTORS OF THE VOLATILITY-AWARE MAP, FROM THE CALMEST (Q1) TO THE MOST VOLATILE (Q5) LOOK-BACK QUINTILE, AVERAGED OVER THE TRAINED MODELS. FACTORS BELOW ONE NARROW THE FORECAST AND FACTORS ABOVE ONE WIDEN IT RELATIVE TO THE UNCALIBRATED FORECAST.

	DiffPTS					MixPTS				
Dataset	Q1	Q2	Q3	Q4	Q5	Q1	Q2	Q3	Q4	Q5
ETTh1	0.55	0.75	0.93	1.02	0.93	0.50	0.77	1.00	1.07	0.97
ETTh2	0.40	0.60	0.72	0.80	0.90	0.40	0.63	0.87	0.90	1.07
ETTm1	0.60	0.75	0.85	1.15	1.47	0.70	0.92	0.97	1.17	1.42
ETTm2	0.40	0.57	0.63	0.68	0.73	0.42	0.75	0.83	0.82	0.85
EXG	0.48	0.87	1.02	1.35	1.65	1.13	1.43	1.38	1.58	1.80
ILI	0.70	0.68	0.73	0.75	0.77	0.75	0.72	0.68	0.82	0.83
Solar	0.97	1.02	1.00	0.90	0.90	1.00	0.93	0.90	0.85	0.80

On SolarEnergy, whose night-time values are exactly zero, the isotonic maps increase the CRPS of both models as well.

I. Seed Ensembles

The ensembles of Section V-G combine the trained models of one method with equal weights and the same total budget of 100 trajectories, drawing 34, 33 and 33 trajectories from the three models (50 and 50 from the two SolarEnergy models of MIXPTS), and they are scored on the dense test windows with the same sampling seed. Table S11 reports all seven datasets. On the four ETT datasets, a single MIXPTS model is more accurate than the three-model DIFFPTS ensemble, with confidence intervals that exclude zero, and the ensemble increases the coverage error of DIFFPTS because it averages

TABLE S10

TEST CRPS AFTER POST-HOC RECALIBRATION FITTED ON THE VALIDATION SPLIT (DENSE TEST WINDOWS; MEANS OVER THE TRAINED MODELS). RAW: NO RECALIBRATION; SCALE: ONE SPREAD FACTOR; VOL.: ONE SPREAD FACTOR PER LOOK-BACK VOLATILITY QUINTILE; ISO: ISOTONIC RECALIBRATION WITH ONE MAP; ISO/VAR: ONE ISOTONIC MAP PER VARIABLE. BOLD MARKS THE BEST VALUE OF EACH ROW.

	DiffPTS					MixPTS				
Dataset	raw	scale	vol.	iso	iso/var	raw	scale	vol.	iso	iso/var
ETTh1	.3530	.3421	.3362	.3421	.3364	.3434	.3368	**.3344**	.3397	.3404
ETTh2	.3692	.3545	.3471	.3433	.3180	.3444	.3390	.3266	.3328	**.3099**
ETTm1	.3129	.3113	.3063	.3108	.3037	.2954	.2962	**.2925**	.2967	.2929
ETTm2	.3046	.2876	.2824	.2749	.2673	.2645	.2594	.2555	.2493	**.2418**
EXG	.2699	.2702	**.2687**	.3287	.3368	.2766	.2867	.2806	.3940	.4247
ILI	**.8395**	.8627	.8595	.8947	.9010	.8573	.8796	.8749	.9061	.9115
Solar	.1868	.1868	.1868	.1892	.1917	.1810	.1811	**.1805**	.2050	.2055

TABLE S11

SEED ENSEMBLES ON THE DENSE TEST WINDOWS. D1/M1: ONE MODEL OF DIFFPTS OR MIXPTS (SCORES AVERAGED OVER THE TRAINED MODELS); D3/M3: EQUALLY WEIGHTED ENSEMBLE OF THE TRAINED MODELS WITH 100 TRAJECTORIES IN TOTAL. THE LAST THREE COLUMNS COMPARE M1 WITH D3: RELATIVE CRPS REDUCTION OF M1, 95% BLOCK-BOOTSTRAP INTERVAL AND SHARE OF WINDOWS ON WHICH M1 HAS THE LOWER CRPS.

	CRPS				ACE				M1 vs. D3		
Dataset	D1	D3	M1	M3	D1	D3	M1	M3	gain (%)	95% CI	win (%)
ETTh1	.3530	.3483	.3434	.3369	.142	.153	.080	.091	+1.4	[0.2, 2.5]	73
ETTh2	.3692	.3641	.3444	.3389	.139	.147	.091	.102	+5.4	[0.2, 11.0]	71
ETTm1	.3129	.3093	.2954	.2908	.058	.066	.036	.045	+4.5	[3.8, 5.2]	87
ETTm2	.3046	.3031	.2645	.2591	.171	.174	.110	.120	+12.7	[11.1, 14.2]	92
EXG	.2699	.2547	.2766	.2548	.049	.062	.168	.112	−8.6	[−16.4, −2.1]	34
ILI	.8395	.7984	.8573	.8080	.246	.188	.287	.216	−7.4	–	23
Solar	.1868	.1805	.1810	.1771	.061	.013	.063	.042	−0.3	[−2.8, 2.4]	44

over-dispersed forecasts. On ExchangeRate, ILI and SolarEnergy, where the single models of both methods are close, the ensemble of DiffPTS is more accurate than one MixPTS model; ensembling MixPTS improves it on every dataset, by 1.5–2.1% on the ETT datasets and SolarEnergy and by 5.7–7.9% on ILI and ExchangeRate.

J. Number of Sampled Trajectories

Table S12 reports the CRPS of the first S trajectories of every dense test window for $S \in \{5, 10, 20, 50, 100\}$, using the trajectories drawn for the analyses of Section V-C; the values at $S = 100$ therefore coincide with Table S4. Since the empirical CRPS of S samples overestimates the CRPS of the predictive law by a term that decays roughly as $1/S$, both methods improve with S; however, MixPTS is more accurate than DiffPTS at every budget and on every dataset, and bold entries mark the budgets at which it already improves on DiffPTS with all 100 trajectories.

TABLE S12

CRPS OF THE FIRST S TRAJECTORIES ON THE DENSE TEST WINDOWS (MEANS OVER THE TRAINED MODELS). BOLD: MIXPTS WITH S TRAJECTORIES IS MORE ACCURATE THAN DIFFPTS WITH ALL 100 TRAJECTORIES.

	DiffPTS, S =					MixPTS, S =				
Dataset	5	10	20	50	100	5	10	20	50	100
ETTh1	.4431	.3957	.3720	.3577	.3530	.4263	.3825	.3607	**.3477**	**.3434**
ETTh2	.4589	.4115	.3880	.3740	.3692	.4245	.3822	**.3611**	**.3486**	**.3444**
ETTm1	.3768	.3431	.3264	.3162	.3129	.3544	.3234	**.3079**	**.2985**	**.2954**
ETTm2	.3835	.3418	.3211	.3088	.3046	.3315	**.2964**	**.2787**	**.2681**	**.2645**
Solar	.2203	.2027	.1938	.1885	.1868	.2189	.1990	.1890	**.1830**	**.1810**

REFERENCES

[1] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," in *Proc. ICLR*, 2023.

[2] Y. Liu, H. Wu, J. Wang, and M. Long, "Non-stationary transformers: Exploring the stationarity in time series forecasting," in *Proc. NeurIPS*, 2022.

[3] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.

[4] W. Ye, D. Li, H. Liu, H. Jiang, Y. Sekimoto, and R. Jiang, "DiffPTS: Rethinking diffusion ELBO for probabilistic time series forecasting," *arXiv:2609.32363*, 2026.

[5] W. Ye, Z. Xu, and N. Gui, "Non-stationary diffusion for probabilistic time series forecasting," in *Proc. ICML*, 2025.

[6] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.

[7] E. P. Grimit, T. Gneiting, V. J. Berrocal, and N. A. Johnson, "The continuous ranked probability score for circular variables and its application to mesoscale forecast ensemble verification," *Quarterly Journal of the Royal Meteorological Society*, vol. 132, no. 621C, pp. 2925–2942, 2006.

[8] D. Nguyen-Cong, L. Tran, and T. Kieu, "When denoising hurts: Rethinking the terminal step of diffusion time series forecasters," *arXiv:2608.14067*, 2026.

[9] V. Kuleshov, N. Fenner, and S. Ermon, "Accurate uncertainties for deep learning using calibrated regression," in *Proc. ICML*, 2018.